

Electrostatic Problem of a Charged Disc

1 Statement of the Problem

The potential and capacitance of a thin, charged circular conducting disc (Problem 3.3 in *Classical Electrodynamics* by Jackson [1]) is arguably one of the most difficult problems in the book. For me, the main difficulty arises from the way the problem is described: it states that the charge is distributed over the entire surface of the disc, rather than concentrated on its edge, as one might expect for a perfect conductor. In electrostatic equilibrium, the charge on a conductor resides on its boundary (see Problem 1.1(a)). This apparent contradiction is beautifully explained by Michael Fowler [2], who points out that the migration of charge to the surface of a conductor follows directly from the inverse-square distance dependence of Coulomb's law. Crucially, this argument holds only in three dimensions. Once this issue concerning the charge distribution on the disc is clarified, the problem can be solved straightforwardly using the method described in Section 3.3.

In this note, I present several additional methods for solving this problem. As is common in electrostatic problems with different geometries, I consider two orthogonal coordinate systems: ellipsoidal and oblate spheroidal coordinates. Although the full solution of the Laplace's equation can in principle be obtained by separation of variables in these coordinate systems, I focus only on particular solutions that arise naturally from families of equipotential surfaces.

2 Ellipsoidal Coordinates

A circular disc can be regarded as the limiting case of an oblate spheroid that has been flattened along its symmetry axis. Since an oblate spheroid is a special type of ellipsoid, the disc may therefore be viewed as a degenerate ellipsoid. Consequently, one can solve the electrostatic problem for a general ellipsoid and then obtain the corresponding result for a circular disc by taking an appropriate limiting case.

2.1 Scaled spherical coordinate

An ellipsoid can be described by the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \quad (1)$$

By analogy with spherical coordinates, one may parameterize the Cartesian coordinates as

$$x = ar \sin \theta \cos \varphi, \quad y = br \sin \theta \sin \varphi, \quad z = cr \cos \theta,$$

where $r = 1$, $\theta \in [0, \pi]$, and $\varphi \in [0, 2\pi]$. However, the parameters r , θ , and φ do *not* form an orthogonal coordinate system. The parametrization can be viewed as the mapping

$$\mathbf{r}(r, \theta, \varphi) = (ar \sin \theta \cos \varphi, br \sin \theta \sin \varphi, cr \cos \theta).$$

The corresponding coordinate basis vectors are

$$\begin{aligned}\mathbf{r}_r &= (a \sin \theta \cos \varphi, b \sin \theta \sin \varphi, c \cos \theta), \\ \mathbf{r}_\theta &= (ar \cos \theta \cos \varphi, br \cos \theta \sin \varphi, -cr \sin \theta), \\ \mathbf{r}_\varphi &= (-ar \sin \theta \sin \varphi, br \sin \theta \cos \varphi, 0).\end{aligned}$$

For an orthogonal coordinate system, these vectors must be mutually perpendicular. However, their dot products are generally nonzero. For example,

$$\mathbf{r}_r \cdot \mathbf{r}_\theta = r \sin \theta \cos \theta (a^2 \cos^2 \varphi + b^2 \sin^2 \varphi - c^2),$$

which does not vanish in general. Similar calculations show that

$$\mathbf{r}_r \cdot \mathbf{r}_\varphi \neq 0, \quad \mathbf{r}_\theta \cdot \mathbf{r}_\varphi \neq 0.$$

Therefore, the coordinates (r, θ, φ) are not orthogonal. Only in the special case $a = b = c$, corresponding to a sphere, do these expressions reduce to the standard spherical coordinates, which are orthogonal. Geometrically, the parametrization arises from scaling spherical coordinates by different factors along the x , y , and z directions. Such anisotropic scaling does not preserve angles, so the originally orthogonal spherical grid becomes skewed on the ellipsoid. Non-orthogonal coordinate systems complicate the expression of the Laplace operator. Consequently, the scaled spherical coordinates introduced above are not well suited for solving this problem.

2.2 Ellipsoidal coordinate

Ellipsoidal coordinates form an orthogonal curvilinear coordinate system naturally adapted to problems involving ellipsoids and related quadratic surfaces. They are particularly useful in solving the Laplace equation for boundary-value problems where the boundary is an ellipsoid. Ellipsoidal coordinates (ξ, η, ζ) are defined as the three roots u of

$$\frac{x^2}{a^2 + u} + \frac{y^2}{b^2 + u} + \frac{z^2}{c^2 + u} = 1,$$

where $a > b > c > 0$. For any point (x, y, z) , the three solutions satisfy

$$-a^2 \leq \zeta \leq -b^2 \leq \eta \leq -c^2 \leq \xi.$$

The ellipsoidal coordinates have a clear geometric interpretation. Surfaces of constant ξ satisfy

$$\frac{x^2}{a^2 + \xi} + \frac{y^2}{b^2 + \xi} + \frac{z^2}{c^2 + \xi} = 1$$

represent a family of ellipsoids. Similarly, surfaces of constant η

$$\frac{x^2}{a^2 + \eta} + \frac{y^2}{b^2 + \eta} + \frac{z^2}{c^2 + \eta} = 1$$

correspond to one-sheeted hyperboloids, and surfaces of constant ζ

$$\frac{x^2}{a^2 + \zeta} + \frac{y^2}{b^2 + \zeta} + \frac{z^2}{c^2 + \zeta} = 1$$

represent two-sheeted hyperboloids. All these families of quadric surfaces are confocal, sharing the same focal structure as the ellipsoid given in (1). A point in three-dimensional space can be uniquely specified by the intersection of these three surfaces.

The transformation from ellipsoidal coordinates (ξ, η, ζ) to Cartesian coordinates is

$$x^2 = \frac{(a^2 + \xi)(a^2 + \eta)(a^2 + \zeta)}{(a^2 - b^2)(a^2 - c^2)}, \quad y^2 = \frac{(b^2 + \xi)(b^2 + \eta)(b^2 + \zeta)}{(b^2 - a^2)(b^2 - c^2)}, \quad z^2 = \frac{(c^2 + \xi)(c^2 + \eta)(c^2 + \zeta)}{(c^2 - a^2)(c^2 - b^2)}.$$

For convenience, we introduce the notation

$$R_\xi = (a^2 + \xi)(b^2 + \xi)(c^2 + \xi),$$

and define R_η and R_ζ analogously. In terms of these quantities, the scale factors of the ellipsoidal coordinate system can be written as

$$h_\xi = \frac{1}{2} \sqrt{\frac{(\xi - \eta)(\xi - \zeta)}{R_\xi}}, \quad h_\eta = \frac{1}{2} \sqrt{\frac{(\eta - \xi)(\eta - \zeta)}{R_\eta}}, \quad h_\zeta = \frac{1}{2} \sqrt{\frac{(\zeta - \xi)(\zeta - \eta)}{R_\zeta}}.$$

Using these scale factors, the Laplace operator in ellipsoidal coordinates takes the form

$$\nabla^2 = \frac{1}{h_\xi h_\eta h_\zeta} \left[\frac{\partial}{\partial \xi} \left(\frac{h_\eta h_\zeta}{h_\xi} \frac{\partial}{\partial \xi} \right) + \frac{\partial}{\partial \eta} \left(\frac{h_\xi h_\zeta}{h_\eta} \frac{\partial}{\partial \eta} \right) + \frac{\partial}{\partial \zeta} \left(\frac{h_\xi h_\eta}{h_\zeta} \frac{\partial}{\partial \zeta} \right) \right].$$

2.3 Solving electrostatic problem in ellipsoidal coordinates

In ellipsoidal coordinates, the Laplace equation takes the form

$$\nabla^2 \Phi(\xi, \eta, \zeta) = 0,$$

or equivalently,

$$\frac{1}{h_\xi h_\eta h_\zeta} \left[\frac{\partial}{\partial \xi} \left(\frac{h_\eta h_\zeta}{h_\xi} \frac{\partial \Phi}{\partial \xi} \right) + \frac{\partial}{\partial \eta} \left(\frac{h_\xi h_\zeta}{h_\eta} \frac{\partial \Phi}{\partial \eta} \right) + \frac{\partial}{\partial \zeta} \left(\frac{h_\xi h_\eta}{h_\zeta} \frac{\partial \Phi}{\partial \zeta} \right) \right] = 0.$$

This equation can be solved by the method of separation of variables. The resulting solutions are known as ellipsoidal harmonics, which play a role analogous to spherical harmonics in problems with ellipsoidal symmetry.

However, for an equipotential ellipsoid with $\xi = 0$, we can assume that the solution only depends on ξ , *i.e.*, $\Phi(\xi, \eta, \zeta) \equiv \Phi(\xi)$. Under this assumption, the Laplace equation simplifies to

$$\frac{d}{d\xi} \left(\sqrt{R_\xi} \frac{d\Phi}{d\xi} \right) = 0.$$

The general solution of this equation is

$$\Phi(\xi) = A \int_\xi^{+\infty} \frac{d\xi'}{\sqrt{R_{\xi'}}} = A \int_\xi^{+\infty} \frac{d\xi'}{\sqrt{(a^2 + \xi')(b^2 + \xi')(c^2 + \xi')}},$$

where the constant A is determined by boundary conditions. This solution naturally satisfies $\Phi \rightarrow 0$ as $\xi \rightarrow +\infty$.

For an ellipsoid carrying a total charge Q , the potential at large distances behaves like

$$\Phi \sim \frac{Q}{4\pi\epsilon_0 r}.$$

In ellipsoidal coordinates, $r \rightarrow +\infty$ corresponds to $\xi \rightarrow +\infty$. At large ξ , the factor under the integral behaves as

$$R_{\xi'} \sim (\xi')^3, \quad \text{so that} \quad \sqrt{R_{\xi'}} \sim (\xi')^{3/2}.$$

Therefore, for large ξ , the potential behaves as

$$\Phi(\xi) \sim 2A/\sqrt{\xi} \sim \frac{2A}{r}.$$

Comparing with the known asymptotic form, we find

$$A = \frac{Q}{8\pi\epsilon_0},$$

and the potential of the charged ellipsoid can therefore be written as

$$\Phi(\xi) = \frac{Q}{8\pi\epsilon_0} \int_{\xi}^{+\infty} \frac{d\xi'}{\sqrt{R_{\xi'}}}.$$

On the ellipsoidal surface, where $\xi = 0$, the potential becomes

$$\Phi(0) = \frac{Q}{8\pi\epsilon_0} \int_0^{+\infty} \frac{d\xi}{\sqrt{R_{\xi}}}.$$

From this, the capacitance of the ellipsoid is given by

$$\frac{1}{C} = \frac{1}{8\pi\epsilon_0} \int_0^{+\infty} \frac{d\xi}{\sqrt{R_{\xi}}}.$$

The surface charge density can be determined from the normal derivative of the potential

$$\sigma = -\epsilon_0 \left. \frac{\partial \Phi}{\partial \xi} \right|_{\xi=0} = -\epsilon_0 \left(\frac{1}{h_{\xi}} \frac{\partial \Phi}{\partial \xi} \right)_{\xi=0} = \frac{Q}{4\pi\sqrt{\eta\zeta}}.$$

Notice that on the surface $\xi = 0$, we have

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} = \frac{\eta\zeta}{a^2b^2c^2}.$$

Therefore, the surface charge density can be expressed as

$$\sigma = \frac{Q}{4\pi abc} \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right)^{-1/2}. \quad (2)$$

For an oblate spheroid, where $a = b > c$, the potential and capacitance can be expressed in closed form. The potential is

$$\Phi(\xi) = \frac{Q}{4\pi\epsilon_0\sqrt{a^2 - c^2}} \arctan \sqrt{\frac{a^2 - c^2}{\xi + c^2}},$$

and the capacitance is

$$C = \frac{4\pi\epsilon_0\sqrt{a^2 - c^2}}{\arccos(c/a)}.$$

In the special case of a circular disc, where $a = b$ and $c = 0$, the capacitance reduces to

$$C = 8\epsilon_0 a.$$

Taking the limit in Eq. (2) as $c \rightarrow 0$ and $z \rightarrow 0$, while maintaining the ratio as

$$\frac{z}{c} = \sqrt{1 - \frac{r^2}{a^2}}, \quad r^2 = x^2 + y^2,$$

the surface charge density becomes

$$\sigma = \frac{Q}{4\pi a^2} \left(1 - \frac{r^2}{a^2}\right)^{-1/2}.$$

3 Oblate Spheroidal Coordinates

References

- [1] J.D. Jackson, *Classical Electrodynamics*, 3rd ed. (John Wiley & Sons, 1999).
- [2] Michael Fowler, Field of a Charged Conducting Disc.
- [3] L. Landau and E. Lifshitz, *Electrodynamics of Continuous Media*, 2nd revised ed. (Butterworth-Heinemann Ltd., 1984).